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Film-Driven Brain Topology

A Multi-Scale Mapper Analysis of fMRI Emotional Dynamics

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Introduction

When we watch a film, our emotions do not switch on and off. They build, mix, and fade in a continuous flow. The brain that supports this experience is just as dynamic: its activity moves through a high-dimensional space of states, one per instant of the film. A central question is how to capture this moving picture without discarding most of its detail. Classical methods often start by averaging activity over time or space, which removes exactly the structure we want to study.

This thesis takes a different route, based on Topological Data Analysis [1]. Using the Mapper algorithm [2], we turn the brain activity recorded during a film into a graph that summarises the shape of its trajectory through state space, without fixing time windows or regions in advance. We apply this to the Emo-Film dataset [3], where 30 subjects watched 14 short films inside an MRI scanner, and we attach to every node of each graph the emotional content of the corresponding moments. The result is a large collection of brain networks, studied at different scales and compared across subjects and films.

The picture that emerges is organised by scale. At the global level, the overall shape of the network does not change between rest and film, even though the networks are clearly not random. One scale below, the internal organisation does change: the core of the network becomes deeper and its communities are rearranged during film viewing. At the level of single nodes, a clear pattern appears: the most central, well-connected states of the brain are emotionally neutral, while strong emotional moments live at the periphery of the graph. Finally, comparing whole networks with a network similarity measure [4], we find that the film pulls different brains toward a common shape, while still leaving a faint but measurable individual signature.

The thesis is organised to follow this thread. Chapters 1 and 2 introduce the mathematical tools, Chapter 3 provides an overview of the dataset, and Chapter 4 reviews the computational pipeline to analyze neural signals as annotated graphs. Chapter 5 presents the results across different scales, from global to local ones, and the comparison of whole graph topologies. Finally, Chapter 6 draws the threads together with an outlook to future directions.

Part I

Theoretical Foundations

Chapter 1

Graph Theory and the Mapper Framework

1.1 Fundamental Graph Theory Concepts

This section collects the graph-theoretic definitions used later to describe and compare the networks studied in this work. It is meant as a reference: a reader familiar with network science can skip it and return only when needed. All graphs considered in this work are undirected and unweighted, following the conventions of Latora et al. [5].

1.1.1 Nodes, Edges, and the Graph

Graph. An undirected graph $G = (\mathcal{N}, L)$ consists of a non-empty set $\mathcal{N} = \{n_1, \dots, n_N\}$ of nodes and a set $L = \{l_1, \dots, l_K\}$ of unordered pairs of distinct nodes called edges. The order of the graph is $N = |\mathcal{N}|$ and its size is $K = |L|$.

Node degree. The degree k_i of a node i is the number of edges incident on it. The average degree of the graph is

$$\langle k \rangle = \frac{1}{N} \sum_{i=1}^N k_i = \frac{2K}{N}. \quad (1.1)$$

Degree distribution. The degree distribution is the probability distribution

$$p_k = \frac{N_k}{N}, \quad (1.2)$$

where N_k is the number of nodes of degree k . It summarises the connectivity profile of the network. The distribution that best fits the empirical graphs is tested in Section 5.2.

1.1.2 Paths, Distances, and Diameter

Shortest path and graph distance. A shortest path, or geodesic, between nodes x and y is a walk of minimal length connecting them. The geodesic distance d_{xy} is the number of edges on this path. The distance is symmetric, $d_{xy} = d_{yx}$, and finite only if x and y lie in the same connected component.

Diameter and characteristic path length. The eccentricity of a node i is $e_i = \max_j d_{ij}$. The diameter is the largest eccentricity in the graph,

$$D = \max_{i,j} d_{ij}, \quad (1.3)$$

and the characteristic path length is the mean geodesic distance over all ordered pairs of distinct nodes,

$$L = \frac{1}{N(N-1)} \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N d_{ij}. \quad (1.4)$$

For a disconnected graph both quantities diverge, so they are computed on the Largest Connected Component only [5].

1.1.3 Connected Components and the Largest Connected Component

Connected component and LCC. A connected component containing node i is the maximal set of nodes reachable from i , together with their edges. The Largest Connected Component (LCC) is the component with the most nodes, and the LCC ratio is the fraction of all nodes it contains.

The empirical graphs have a mean LCC ratio of 0.53 (Section 5.2), so several meso-scale analyses are run on the LCC rather than on the full graph.

1.1.4 Graph Density

Graph density. The density of a graph with N nodes and K edges is the ratio of actual to possible edges,

$$\rho = \frac{K}{\binom{N}{2}} = \frac{2K}{N(N-1)}. \quad (1.5)$$

It ranges from $\rho = 0$ for an empty graph to $\rho = 1$ for a complete graph. The empirical graphs are sparse, since the vast majority of node pairs are not connected by an edge. Density is reported for all 420 empirical graphs in Section 5.2.

1.2 Network Meso-Scale Metrics: K-core Decomposition and Betweenness Centrality

Global metrics describe a network as a whole but say little about its internal organisation. Two meso-scale metrics fill this gap: k -core decomposition, which finds nested subgraphs of increasing cohesion, and betweenness centrality, which measures how much a node acts as an intermediary. Both are reported for the empirical graphs in Section 5.3, and are later combined into the node classification defined in Section 4.3.

1.2.1 K-core Decomposition

K-core. The k -core of a graph G is the largest induced subgraph $H \subseteq G$ in which every node has degree at least k within H : $\deg_H(i) \geq k$ for all $i \in H$.

The k -core is obtained by repeatedly removing all nodes of degree below k and recomputing degrees until none remain [6]. This produces a nested sequence $H_k \subseteq H_{k-1} \subseteq \dots \subseteq H_1 \subseteq G$.

Coreness / shell index. The coreness c_i of a node i is the highest k for which i belongs to the k -core:

$$c_i = \max\{k : i \in H_k\}. \quad (1.6)$$

The nodes with coreness exactly k form the k -shell, and $c_{\max} = \max_i c_i$ measures the depth of the core hierarchy.

A node with high coreness sits in a dense, mutually cohesive core of the network. In the empirical graphs the coreness ranges from 0 to 4, with a mean maximum coreness of 3.25 (Section 5.3).

1.2.2 Betweenness Centrality

Betweenness centrality. The betweenness centrality of a node i is [5]

$$c_i^B = \sum_{j \neq i} \sum_{k \neq i, j} \frac{n_{jk}(i)}{n_{jk}}, \quad (1.7)$$

where n_{jk} is the number of geodesics from j to k and $n_{jk}(i)$ the number of those passing through i . The normalised version is

$$C_i^B = \frac{c_i^B}{(N-1)(N-2)}, \quad (1.8)$$

which lies in $[0, 1]$ and equals 1 when i lies on every geodesic between other pairs.

A node with high betweenness bridges otherwise distant parts of the network, lying on many of the shortest paths that connect them. In the empirical graphs

only 22.8% of nodes have non-zero betweenness, a consequence of the sparse, locally clustered topology (Section 5.3).

1.3 Modularity Optimization and the Louvain Community Detection Algorithm

Many networks are not homogeneous: their nodes tend to organise into subgroups whose members are more tightly linked to each other than to the rest of the graph. These subgroups are called communities, and community detection is the problem of recovering them from the network structure alone. A good partition is one where most edges fall inside communities and few fall between them. This section defines modularity, the function that measures how good a partition is, and the Louvain algorithm that maximises it. Both are applied to the empirical empirical graphs, restricted to the Largest Connected Component (Section 5.3).

1.3.1 Modularity

Modularity. For a graph G with K edges and a partition $P_N = \{C_1, \dots, C_M\}$ of its nodes, the modularity is [5]

$$Q = \sum_{m=1}^M \left[\frac{K_{mm}}{K} - \left(\frac{K_m}{2K} \right)^2 \right], \quad (1.9)$$

where K_{mm} is the number of edges inside community C_m and K_m is the sum of the degrees of its nodes. An equivalent form over node pairs is [5]

$$Q = \frac{1}{2K} \sum_{i=1}^N \sum_{j=1}^N \left[a_{ij} - \frac{k_i k_j}{2K} \right] \delta(C(i), C(j)), \quad (1.10)$$

where a_{ij} is the adjacency matrix entry and $\delta(C(i), C(j)) = 1$ if i and j share a community, 0 otherwise.

The first term is the actual fraction of edges inside each community, the second the fraction expected in a random graph with the same degree sequence. Modularity is therefore high when communities hold more internal edges than chance would give. Values near 0 mean no structure beyond random, larger values mean well-separated modules, and Q can be negative for a partition worse than random.

1.3.2 The Louvain Algorithm

Exact modularity maximisation is computationally intractable for large graphs. The Louvain algorithm is an efficient heuristic that alternates two phases [7].

In the first phase every node starts in its own community. Each node i is then moved to the neighbouring community that gives the largest positive modularity gain

$$\Delta Q = \frac{K_{i,\text{in}}}{K} - \frac{\Sigma_{\text{tot}} \cdot k_i}{2K^2}, \quad (1.11)$$

where $K_{i,\text{in}}$ is the edge weight between i and the community, Σ_{tot} the total weight incident to it, and k_i the degree of i . If no move improves Q , the node stays. This repeats over all nodes until a local maximum is reached.

In the second phase each community is collapsed into a single node, with edge weights summed and internal edges kept as self-loops. The two phases together form one pass, and the procedure repeats on the coarser graph until Q can no longer increase. The result is a hierarchical decomposition of the network.

In this work Louvain is run with the `python-louvain` library using default parameters (no resolution scaling), on the Largest Connected Component of each graph. The restriction is necessary because the null term $k_i k_j / 2K$ is not well defined across disconnected components. The comparison against null models is reported in Section 5.3.

1.4 Shannon Entropy and Information-Theoretic Graph Descriptors

Comparing two graphs is not trivial: they may have the same number of nodes and edges yet differ in their topology at several scales. This comparison is central to the fingerprinting analysis of Section 5.5, where graphs from different subjects and films are compared to weigh the role of individual identity against that of the film. For this we use the Network Portrait Divergence of Bagrow and Boltt [4], which measures structural differences at all scales at once and needs no node-to-node correspondence between the two graphs.

1.4.1 The Graph Portrait

Graph Portrait. For an undirected graph G with N nodes and diameter d , the network portrait is the matrix B with entries [4]

$$B_{\ell,k} = \text{number of nodes with exactly } k \text{ nodes at geodesic distance } \ell, \quad (1.12)$$

for $0 \leq \ell \leq d$ and $0 \leq k \leq N - 1$.

The portrait is a graph invariant: it is identical for isomorphic graphs and does not depend on how the nodes are labelled. Each row captures a different scale. Row $\ell = 1$ gives $B_{1,k} = N \cdot p_k$, the degree distribution; row $\ell = 2$ gives the distribution of next-nearest neighbours, and so on up to the diameter. A single matrix therefore holds the local, meso-scale, and global structure of the graph.

1.4.2 The Portrait Divergence

To compare two graphs G and G' with portraits B and B' , each portrait is turned into a probability distribution. The probability of picking two connected nodes at distance ℓ , one of which has exactly k nodes at that distance, is [4]

$$P(k, \ell) = \frac{kB_{\ell,k}}{\sum_c n_c^2}, \quad (1.13)$$

where n_c are the sizes of the connected components. The distribution $Q(k, \ell)$ is built the same way from B' .

Network Portrait Divergence. The Network Portrait Divergence between G and G' is the Jensen-Shannon divergence of their distributions [4]

$$D_{\text{JS}}(G, G') = \frac{1}{2}\text{KL}(P \parallel M) + \frac{1}{2}\text{KL}(Q \parallel M), \quad (1.14)$$

where $M = \frac{1}{2}(P + Q)$ and KL is the Kullback-Leibler divergence.

This measure lies in $[0, 1]$, is symmetric, and its square root is a valid metric. A value of 0 means the two portraits are identical, and larger values mean greater topological difference. Because $P(k, \ell)$ combines all distance shells, the divergence is scale-free: it is sensitive to differences from local degree patterns up to global path lengths.

In Section 5.5 the divergence is computed between all pairs of empirical graphs to form two sets of distances: within-subject (same subject, different films) and within-film (same film, different subjects). Comparing the two distributions gives a direct test of whether brain topology is driven more by the film or by the identity of the subject.

1.5 The Mapper Algorithmic Framework

Naturalistic neuroimaging produces high-dimensional data, and classical approaches such as pairwise correlations or linear dimensionality reduction collapse information across space or time before the analysis even starts. The Mapper algorithm, introduced by Singh, Mémoli, and Carlsson [2], takes a different route. Instead of looking for a low-dimensional embedding, it builds a graph that captures the shape of the data at a chosen resolution. The idea behind it is that the shape of the data cloud carries information that averaging or projection destroys. Two regions of the space with similar mean activation but different geometry look identical once averaged, yet they trace different paths through the high-dimensional space.

This view fits fMRI data well. Each time frame, or TR, is a point in a space with one dimension per brain region, and the sequence of TRs is a continuous

trajectory through that space. As Saggar et al. [1] note, classical time-varying analyses lose information because they fix time windows or average over space from the start. Mapper instead summarises the brain’s dynamics without collapsing the data in space or time, keeping the full resolution of the acquisition [1].

1.5.1 The Algorithmic Pipeline

Applied to fMRI data [1, 8], Mapper runs in four steps, summarised in Figure 1.1.

Step 1 – Filter function: The input $X \in \mathbb{R}^{N \times d}$, with N TRs and d brain regions, is mapped into a lower-dimensional space by a filter function $f : X \rightarrow \mathbb{R}^k$, with $k \ll d$. The filter organises the data along a few axes before clustering. Any function that reflects meaningful structure is admissible [2]. We use two, Reciprocal Isomap (R-Isomap) and t -SNE, each giving a different geometric view of the data (Section 1.6).

Step 2 – Binning: The range of the filter is split into overlapping bins. Two parameters control this: the resolution R , the number of bins, and the gain G , the overlap between adjacent bins. A finer resolution gives more and smaller nodes, a larger gain gives more edges. The overlap is essential: when two adjacent bins share points, the clusters drawn from them can intersect, and that is what creates the edges of the final graph.

Step 3 – Partial clustering: Inside each bin, the TRs falling in that bin are clustered in the original high-dimensional space, not in the projected one. Each cluster becomes a node. The clustering is partial because it runs on overlapping subsets rather than on the whole dataset, and because it works on the full-dimensional data, it recovers geometric relations that the filter projection had compressed [8]. Following Saggar et al. [1], we use single-linkage clustering with Manhattan (L_1) distance.

Step 4 – Graph construction: Each cluster is a node, and an edge joins two nodes whose clusters share at least one TR. Node size is proportional to the number of TRs in the cluster.

1.5.2 Interpretation in the fMRI Context

The result, called the shape graph [1, 8], has a natural reading. Each node is a set of TRs with similar whole-brain activation, and an edge means at least one TR is shared between two clusters, marking a transition between the two brain states. The graph thus traces the evolution of whole-brain states over time, with no fixed segmentation into time windows.

Saggar et al. [8] showed that on resting-state data this representation has a clear structure: central, high-degree nodes are states visited often and act as transitions between resting-state networks, while peripheral nodes are dominated by one or few networks. This thesis applies the same logic to emotional film stimuli. Each node is annotated with the emotional content of its TRs (Section 4.4), which links the position of a node in the graph to the emotion of the corresponding brain states. This link between topology and emotion is the core of Chapter 5.

1.5.3 Resolution, Gain, and Multi-Scale Analysis

The resolution R and gain G set the granularity of the graph and must be chosen carefully. Saggar et al. [1] tested the robustness of their approach by varying R from 10 to 70 and G from 2 to 5, and found that the main mesoscale properties of the shape graphs held across most combinations.

Here, graphs are generated in seven configurations (Section 4.2) that vary both the parcellation resolution (100, 200, 400, 800 regions) and the filter function (R -Isomap and t -SNE), giving 420 graphs per configuration. This design allows us to check the cross-resolution consistency of the results as an explicit robustness test (Section 5.3).

1.6 Dimensionality Reduction Projection Functions: R-Isomap and t-SNE

The first step of the Mapper pipeline (Section 1.5) applies a filter function $f : \mathbb{R}^d \rightarrow \mathbb{R}^k$ that projects the data into a lower-dimensional space before binning and clustering. We use two filter functions, Reciprocal Isomap (R-Isomap) and t-SNE. Both are non-linear methods, so they can capture curved, low-dimensional structure that linear techniques such as PCA or classical MDS cannot.

1.6.1 Reciprocal Isomap (R-Isomap)

R-Isomap extends the classical Isomap framework [9] by changing how the neighbourhood graph is built, which makes it more robust to outliers.

Reciprocal k -NN graph. Given a dataset $X = \{x_1, \dots, x_N\}$ and an integer k , the reciprocal k -nearest-neighbour graph has an edge between x_i and x_j only if each is among the k nearest neighbours of the other.

Unlike a standard k -NN graph, which adds an edge whenever x_j is a neighbour of x_i regardless of the reverse, the reciprocal rule enforces symmetry and removes the one-sided links that outliers and uneven density tend to create [10].

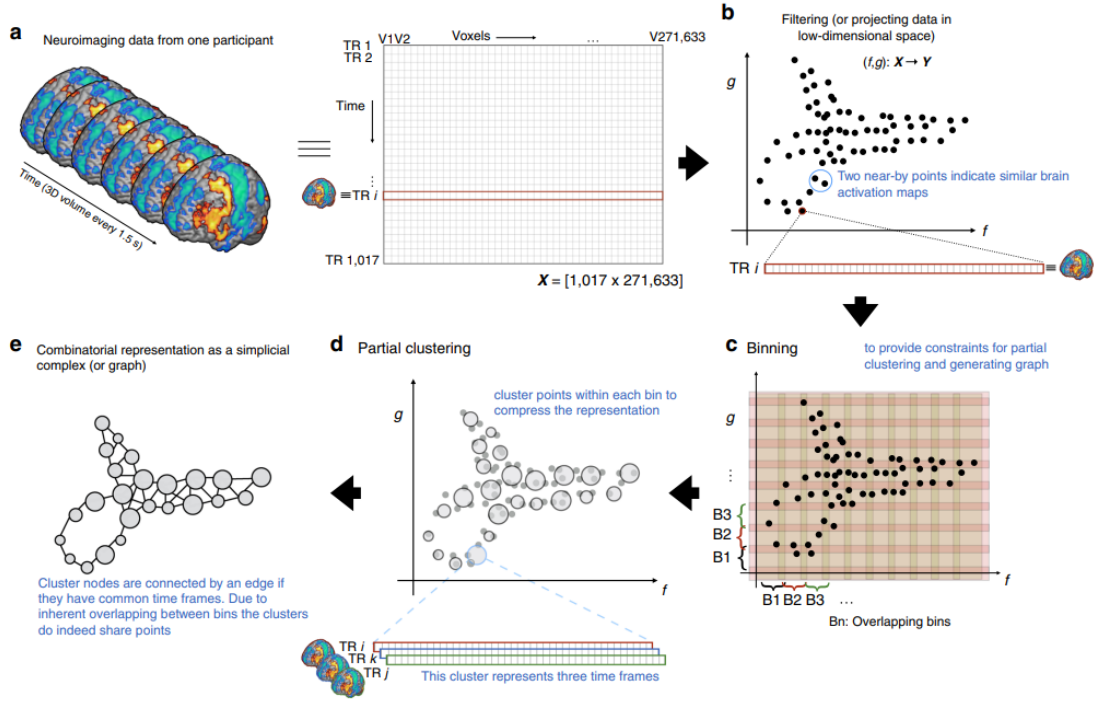


Figure 1.1: Overview of the Mapper algorithmic pipeline. (a) Pre-processed fMRI timeseries are structured as a high-dimensional matrix of timeframes (TRs) across voxels or regions of interest. (b) A non-linear dimensionality reduction step projects the high-dimensional brain states into a lower-dimensional filter space. (c) The low-dimensional filter output is divided into overlapping bins governed by resolution and gain parameters. (d) Partial clustering is performed within each bin in the original high-dimensional space to identify clusters of similar brain states. (e) Nodes representing overlapping clusters are linked by an edge to construct the final combinatorial shape graph. Adapted from Saggar et al. [1].

R-Isomap then estimates the geodesic distance between all pairs as the shortest-path distance on this graph. For nearby points the Euclidean distance is a good proxy, while for distant points chaining short hops along the graph approximates the true distance along the data manifold far better than a straight line [9]. The resulting distance matrix is embedded into a low-dimensional space by classical MDS, and this embedding is the filter output. The key property of R-Isomap is that it preserves global geometry: points far apart along the manifold stay far apart in the embedding, and nearby points stay close.

1.6.2 t-SNE

t-SNE [11] takes a different approach, modelling pairwise similarities as probabilities rather than geodesic distances.

t-SNE similarity model. For each pair of points x_i and x_j , t-SNE defines the joint probability

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad p_{j|i} = \frac{\exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma_i^2}\right)}{\sum_{k \neq i} \exp\left(-\frac{\|x_i - x_k\|^2}{2\sigma_i^2}\right)}, \quad (1.15)$$

where σ_i is the bandwidth of the Gaussian kernel at x_i , set so that the perplexity $\text{Perp}(P_i) = 2^{H(P_i)}$ matches a chosen value, with $H(P_i)$ the Shannon entropy of P_i .

In the low-dimensional space the similarities between the map points y_i and y_j use a Student t -distribution with one degree of freedom,

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_k \sum_{m \neq k} (1 + \|y_k - y_m\|^2)^{-1}}, \quad (1.16)$$

and the embedding is found by minimising the Kullback-Leibler divergence between the two distributions,

$$C = \sum_i \sum_{j \neq i} p_{ij} \log \frac{p_{ij}}{q_{ij}}. \quad (1.17)$$

The heavy tails of the Student t -distribution address the crowding problem, as they let distant clusters sit further apart in the map and stop separate groups from collapsing toward the centre. The perplexity sets how many neighbours each point effectively considers, which balances local separation against global arrangement.

1.6.3 Role as Filter Functions

Both methods act as the filter function f on which the binning step operates, and they offer two different views of the data. R-Isomap arranges the TRs by their

geodesic distances along the manifold, favouring global structure. t-SNE arranges them by local similarity, favouring the separation of distinct states. Using both across the seven configurations (Section 4.2) lets us check that the downstream findings do not depend on a single projection (Section 5.3).

Chapter 2

Statistical Methods and Null Models

2.1 Structural Null Models: Erdős-Rényi and Barabási-Albert

A null model is a random graph used as a statistical baseline. Given an empirical network, the null model produces random graphs that keep a few global properties, such as the node count and edge count, while randomising everything else. Comparing a metric measured on the empirical graph against its distribution over many null graphs tells us whether an observed property is unusual or just a by-product of the network's size and density.

We use two null models with very different degree distributions, so that each one tests a different hypothesis about the Mapper graphs.

2.1.1 The Erdős-Rényi Random Graph $G(n, p)$

The Erdős-Rényi (ER) model is the simplest random graph: every pair of nodes is connected with the same probability, independently of all others. We use the binomial form $G(n, p)$ [5].

Erdős-Rényi random graph $G(n, p)$. Let $n \geq 1$ be the number of nodes and $p \in [0, 1]$ the connection probability. The ensemble $G(n, p)$ contains all undirected simple graphs on n nodes in which each pair $\{i, j\}$ is connected independently with probability p . A graph with K edges has probability

$$\mathbb{P}(G) = p^K (1 - p)^{\binom{n}{2} - K}. \quad (2.1)$$

Since edges are placed independently, the degree of a node follows a binomial distribution with parameters $n - 1$ and p . In the sparse limit, where $n \rightarrow \infty$ and

$p \rightarrow 0$ with the mean degree $\langle k \rangle = p(n-1) = z$ fixed, this converges to a Poisson distribution [5]:

$$p_k = e^{-z} \frac{z^k}{k!}, \quad k = 0, 1, 2, \dots \quad (2.2)$$

The Poisson distribution is concentrated around its mean, with standard deviation $\sqrt{z}$. Almost all nodes have a degree close to the average, and high-degree hubs are very rare. The expected clustering coefficient equals p , which is tiny in sparse graphs. ER graphs are therefore structurally homogeneous: no hubs, no preferential wiring, and no built-in community structure.

This makes $G(n, p)$ the standard baseline null. It encodes the hypothesis that the empirical network has no organisation beyond its edge density. To match an empirical graph with n nodes and $|E|$ edges, the connection probability is set to its maximum likelihood estimate [5]:

$$\hat{p} = \frac{2|E|}{n(n-1)}. \quad (2.3)$$

2.1.2 The Barabási-Albert Scale-Free Model

The Barabási-Albert (BA) model was introduced to explain the heavy-tailed, power-law degree distributions seen in many real networks. Unlike ER, where edges are placed uniformly, BA grows the graph over time and lets well-connected nodes attract still more connections [5].

Barabási-Albert model. Given integers N , n_0 , and m with $m \leq n_0$, the graph is built as follows. At $t = 0$ it starts as a complete graph on n_0 nodes. At each step $t = 1, \dots, N - n_0$:

1. a new node is added with m edges;
2. each edge joins the new node to an existing node i , chosen with probability proportional to its current degree,

$$\Pi(k_i) = \frac{k_i}{\sum_j k_j} = \frac{k_i}{2l_{t-1}}, \quad (2.4)$$

where l_{t-1} is the number of edges before the new node arrives.

The process stops when the graph reaches N nodes.

The rule $\Pi(k_i) \propto k_i$ is called linear preferential attachment: a node gains new links at a rate proportional to its current degree, so early hubs keep growing. The resulting degree distribution can be derived analytically [5] and is

$$p_k = \frac{2m(m+1)}{k(k+1)(k+2)}, \quad (2.5)$$

which decays as $p_k \sim k^{-3}$ for large k , an exponent $\gamma = 3$ independent of m .

A network whose degree distribution follows $p_k \sim k^{-\gamma}$ is called scale-free, because the power law has no characteristic degree scale. In practice this means many low-degree nodes coexist with a few large hubs that carry a large share of the edges, a feature ER graphs do not have. This is why BA is the more informative null for the heavy-tailed degree distribution of the Mapper graphs, where ER would be a poor comparison. Figure 2.1 contrasts an empirical Mapper graph with ER and BA null models matched in node count and density.

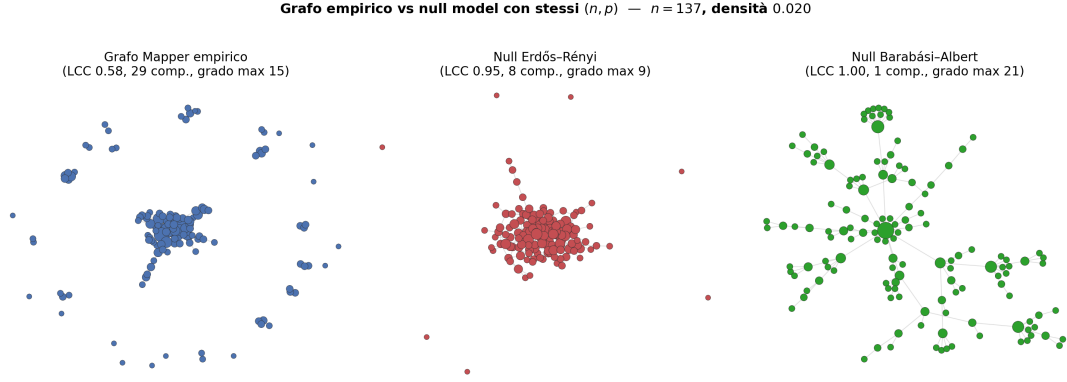


Figure 2.1: Qualitative comparison between an empirical Mapper graph (film *Sintel*, subject S08, configuration 100_riso) and two structural null models matched in node count and edge density. Node size is proportional to degree, mapped on a shared scale across panels for strict comparability. The Erdős–Rényi model generates a structurally homogeneous topology completely devoid of hubs. The Barabási–Albert model yields a fully connected network dominated by a small set of prominent high-degree hubs. Conversely, the empirical brain network exhibits striking fragmentation (its largest connected component spanning 58% of the total nodes) and lacks highly pronounced hubs.

2.1.3 Substrate Matching and Benchmarking Strategy

For each empirical Mapper graph, both null models are matched to the same node count n and edge count $|E|$. For ER this is done by fixing $\hat{p}$ as above. For BA the parameter m is set to the integer closest to $|E|/n$, since a BA graph has about $m \cdot n$ edges, giving a graph with n nodes and approximately $|E|$ edges.

A second choice concerns the substrate on which the null is built. Several metrics in this work are defined only on connected graphs and are computed on the Largest Connected Component (Section 1.1), while others are properties of the full, possibly fragmented graph. To keep every comparison fair, each null is matched to the same substrate as the empirical metric it is compared to:

- metrics defined on the full graph, such as the LCC ratio and the maximum degree, are compared against nulls built on the full graph with n nodes;
- metrics defined on the connected core, such as diameter, modularity, and betweenness, are compared against nulls built on the LCC, matched to its node and edge count.

This single rule, matching the null to the substrate of the metric, ensures that like is always compared with like. In particular, building the null on the full graph for the LCC ratio is what lets us test the network’s fragmentation, since an ER or BA graph fragments to its own degree.

The two models are complementary. ER tests whether a metric is explained by random wiring at the observed density. BA tests whether it is explained by scale-free, preferential-attachment growth. A property that departs from both nulls cannot be attributed to density, random wiring, or degree heterogeneity alone, and points to higher-order structure specific to the brain data. The results of these comparisons are reported in Sections 5.2 and 5.3.

2.2 Degree Distribution Analysis and Power-Law Testing

The shape of the degree distribution says a lot about a network’s structural regime. We compare four candidate forms for the degree sequence of the Mapper graphs: the Poisson distribution (typical of ER graphs), the pure power law, the truncated power law (a power law with an exponential cutoff), and the exponential distribution. For a discrete variable $k \geq x_{\min}$ they are, respectively,

$$p_k = e^{-z} \frac{z^k}{k!}, \quad (2.6)$$

$$p_k \propto k^{-\alpha}, \quad (2.7)$$

$$p_k \propto k^{-\alpha} e^{-\lambda k}, \quad (2.8)$$

$$p_k \propto e^{-\lambda k}. \quad (2.9)$$

The Poisson form has a single scale $z = \langle k \rangle$. The two power-law forms are heavy-tailed, with an exponent $\alpha > 1$ that controls how slowly the tail decays. The exponential form decays faster than any power law and has no heavy tail. Figure 2.2 shows the four candidate shapes as complementary cumulative distribution functions on log–log axes. To decide which model fits best, we follow the framework of Clauset, Shalizi, and Newman [12], which combines maximum likelihood estimation, a bootstrap goodness-of-fit test, and a likelihood ratio comparison between models.

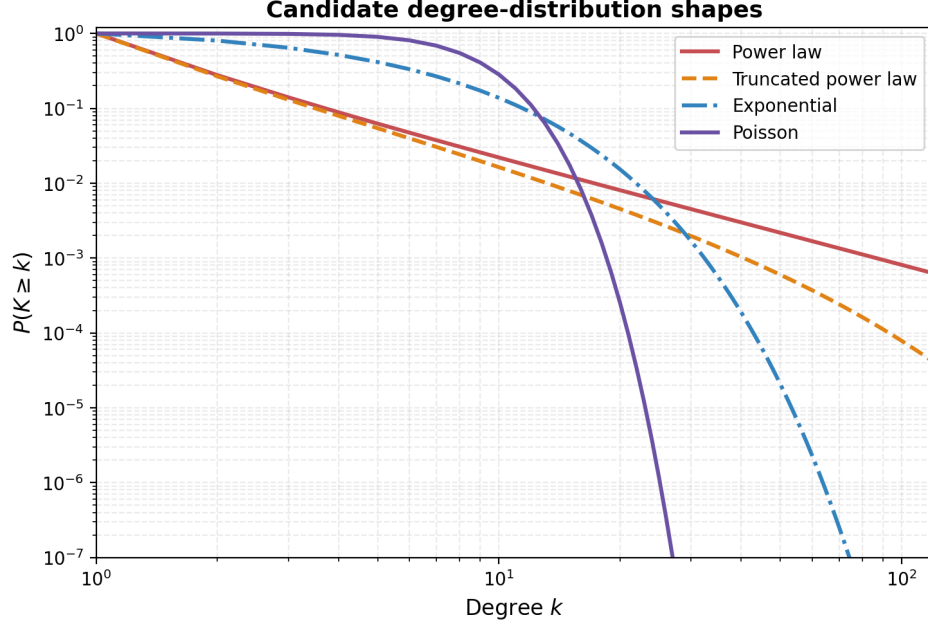


Figure 2.2: The four candidate degree-distribution shapes of Equations (2.6)–(2.9), displayed as complementary cumulative distribution functions (CCDFs), $P(K \geq k)$, on log–log axes to directly compare the geometric decay of the tails. The pure power law (Equation (2.7)) manifests as a straight line, representing the slowest-decaying tail. The truncated power law (Equation (2.8)) tracks the pure power law initially and subsequently bends downwards as the exponential cutoff takes effect. The exponential distribution (Equation (2.9)) decays faster, lacking a heavy-tailed regime altogether. The Poisson distribution (Equation (2.6)), characteristic of Erdős–Rényi random graphs, remains flat up to its mean and then falls off sharply, constituting the lightest tail of the four models. All curves are analytical, plotted with an exponent $\alpha = 2.4$ and representative scale parameters selected for illustrative clarity.

2.2.1 Maximum Likelihood Estimation of α and $x_{\min}$

Fitting the exponent by a straight line on a log-log histogram is known to be biased and should be avoided [12]. Maximum likelihood estimation (MLE) gives unbiased and consistent estimates instead.

For a discrete power law with lower bound $x_{\min}$, the MLE of α solves [12]

$$-\frac{\zeta'(\hat{\alpha}, x_{\min})}{\zeta(\hat{\alpha}, x_{\min})} = \frac{1}{n} \sum_{i=1}^n \ln x_i, \quad (2.10)$$

where $\zeta(\alpha, x_{\min}) = \sum_{k=x_{\min}}^{\infty} k^{-\alpha}$ is the Hurwitz zeta function and the prime denotes

the derivative with respect to α . A closed-form approximation, accurate when $x_{\min} \gtrsim 6$, is

$$\hat{\alpha} \approx 1 + n \left[\sum_{i=1}^n \ln \frac{x_i}{x_{\min} - \frac{1}{2}} \right]^{-1}, \quad (2.11)$$

with standard error $\sigma \approx (\hat{\alpha} - 1)/\sqrt{n}$ [12].

The lower bound $x_{\min}$ is not fixed in advance but estimated from the data, by choosing the value that minimises the Kolmogorov-Smirnov (KS) distance between the empirical distribution and the fitted power law above $x_{\min}$ [12]. Only the observations with $k \geq x_{\min}$ enter the fit, while those below are kept for the goodness-of-fit step.

2.2.2 Bootstrap Goodness-of-Fit

A good MLE fit does not by itself prove the data follow a power law. To test this, Clauset et al. [12] use a bootstrap based on the KS statistic. Let D be the KS distance between the empirical distribution and the fitted power law above $x_{\min}$. The p -value is computed as follows:

1. Generate many synthetic datasets. Each point is drawn from the fitted power law with probability n_{tail}/n , where n_{tail} is the number of empirical points above $x_{\min}$, and sampled from the empirical points below $x_{\min}$ otherwise.
2. For each synthetic dataset, re-estimate $\hat{\alpha}$ and $x_{\min}$ and compute its own KS distance D^* .
3. The bootstrap p -value is the fraction of synthetic datasets with $D^* > D$.

A large p -value (close to 1) means the difference between the data and the power law is consistent with sampling noise. Following the criterion of Clauset et al. [12], the power-law hypothesis is rejected when $p \leq 0.1$.

2.2.3 Likelihood Ratio Test for Model Selection

When the goodness-of-fit test does not reject the power law, we still need to check whether another distribution fits as well or better. For this, Clauset et al. [12] use the Vuong likelihood ratio test, which compares two models directly without assuming that either is the true distribution.

Given two candidates with log-likelihoods ℓ_1 and ℓ_2 evaluated on the data above $x_{\min}$, the log-likelihood ratio is

$$\mathcal{R} = \ell_1 - \ell_2. \quad (2.12)$$

The sign of $\mathcal{R}$ shows which model fits better: $\mathcal{R} > 0$ favours the first, $\mathcal{R} < 0$ the second. Since $\mathcal{R}$ is a random quantity for finite samples, Vuong's test normalises it to a standard normal statistic whose p -value tells whether the preference is significant. A small p -value ($p < 0.05$) means a clear preference, a large one means the two models cannot be told apart.

2.3 Non-Parametric Hypothesis Testing

The graph metrics in this study come from a small sample of $n = 30$ subjects and are generally not normally distributed: Shapiro-Wilk tests confirmed non-normality for the main metrics. We therefore use rank-based non-parametric tests, which make no distributional assumption beyond continuity and are valid at these sample sizes.

2.3.1 Wilcoxon Signed-Rank Test

The Wilcoxon signed-rank test is a paired test for whether the differences between two matched samples are symmetric around zero [13]. Given n pairs (X_{1i}, X_{2i}) , one computes the differences $D_i = X_{1i} - X_{2i}$, drops the zeros, and ranks the absolute values $|D_i|$ from 1 to n' . The statistic is

$$W^+ = \sum_{i: D_i > 0} R_i^+, \quad (2.13)$$

where R_i^+ is the rank of $|D_i|$. Under the null hypothesis of symmetry around zero, W^+ has a known distribution that approaches normality for large n' [13]. We use this test for all paired within-subject contrasts, such as real versus rest graphs and real versus null graphs, with a two-sided alternative.

2.3.2 Mann-Whitney U Test

The Mann-Whitney U test compares two independent samples, testing whether a random value from one is equally likely to exceed a random value from the other [14]. The $N = n_1 + n_2$ observations are pooled and ranked. With R_1 the rank sum of the first sample,

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1, \quad U_2 = n_1 n_2 - U_1, \quad (2.14)$$

and $U = \min(U_1, U_2)$. For moderate sizes, U is approximately normal with mean $n_1 n_2 / 2$ and variance

$$\sigma_U^2 = \frac{n_1 n_2 (N + 1)}{12}. \quad (2.15)$$

We use this test to compare the within-subject and within-film blocks of the portrait divergence matrices.

2.3.3 Kruskal-Wallis Test

The Kruskal-Wallis test is the non-parametric version of one-way ANOVA, testing whether $k \geq 2$ independent groups come from the same distribution [15]. The $N = \sum_{j=1}^k n_j$ observations are pooled and ranked, and with R_j the rank sum of group j ,

$$H = \frac{12}{N(N+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} - 3(N+1). \quad (2.16)$$

Under the null hypothesis, H is approximately χ^2 with $k - 1$ degrees of freedom. We use it to test whether emotional activation and complexity differ across the four node categories (Peripheral, Bridge, Connector Hub, and Provincial Hub) defined in Section 4.3.

2.3.4 Spearman Rank Correlation

Spearman's coefficient measures the strength and direction of a monotonic relation between two variables, without assuming linearity or normality [16]. Each variable is replaced by its rank, and

$$\rho_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}, \quad (2.17)$$

where $d_i = \text{rank}(x_i) - \text{rank}(y_i)$. The coefficient lies in $[-1, 1]$, with ± 1 a perfect monotonic relation. Significance uses the t -approximation

$$t = \rho_s \sqrt{\frac{n-2}{1-\rho_s^2}}, \quad (2.18)$$

which follows a t -distribution with $n-2$ degrees of freedom under the null hypothesis of no association. We use Spearman correlation to relate nodal metrics (degree, betweenness, k -core number) to the emotional profiles across the 44,197 nodes, and to measure cross-resolution consistency.

2.3.5 Multiple Comparisons Correction

When many tests are run at once, the chance of at least one false positive under the global null grows with the number of tests s . We use two corrections.

The *Bonferroni correction* controls the Family-Wise Error Rate (FWER), the probability of one or more false rejections across the whole family [13]. A hypothesis H_i is rejected when

$$\hat{p}_i \leq \frac{\alpha}{s}, \quad (2.19)$$

which guarantees $\text{FWER} \leq \alpha$ regardless of how the tests are correlated [13].

The *Benjamini-Hochberg procedure* controls the False Discovery Rate (FDR), the expected fraction of false positives among the rejected hypotheses [17]. With the p -values sorted as $\hat{p}_{(1)} \leq \dots \leq \hat{p}_{(s)}$, it rejects $H_{(1)}, \dots, H_{(k)}$ with

$$k = \max \left\{ i : \hat{p}_{(i)} \leq \frac{i}{s} \alpha \right\}, \quad (2.20)$$

controlling the FDR at level α under independence [17]. The FDR criterion is less strict than FWER and keeps more power when s is large.

In this work the Bonferroni correction and the Benjamini-Hochberg procedure are both applied to the per-pair film comparisons, where the number of simultaneous tests is large. The specific use of each correction is reported with the corresponding results in Chapter 5.

Part II

**Experimental Framework
and Application**

Chapter 3

Computational Neuroscience: Data and Dataset

3.1 Functional Neuroimaging: BOLD Signal and ROI Timeseries

Functional Magnetic Resonance Imaging (fMRI) measures brain activity indirectly, exploiting the fact that active brain regions consume oxygen at a different rate than they receive it. When neurons fire, the local blood supply responds by delivering more oxygenated blood than is immediately needed. This shift in the ratio between oxygenated and deoxygenated haemoglobin changes the local magnetic properties of the tissue: deoxyhaemoglobin weakly distorts the magnetic field in its surroundings, while oxyhaemoglobin does not. An MRI scanner tuned to detect these distortions therefore picks up a signal that tracks, with a short delay, the underlying pattern of neural activity. This signal is known as the Blood-Oxygen-Level-Dependent (BOLD) contrast [18].

In the Emo-FilM dataset, functional volumes were acquired on a 3 T scanner with a Repetition Time (TR) of 1.3 s [3]: one whole-brain volume every 1.3 seconds, for a total of 460 to 945 TRs per film run depending on duration. Prior to analysis, the data underwent standard preprocessing: head motion correction, alignment to a common reference space, spatial smoothing, temporal filtering, and regression of physiological noise [3].

The pipeline described in this thesis does not operate on individual voxels. After parcellating the cortex into regions of interest using the Schaefer atlas [19] (Section 3.2), the BOLD signal within each region is averaged across all its constituent voxels, yielding one representative timeseries per region per run. Each TR is thereby represented as a single point in an n -dimensional space, where n is the

number of parcellations, and the full run traces a continuous trajectory through that space. This sequence of region-averaged vectors is the direct input to the Mapper algorithm described in Section 1.5.

3.2 Brain Parcellation: The Schaefer Cortical Atlas

Working directly with the tens of thousands of voxels that make up the cortical surface is computationally intractable. Brain parcellation addresses this by grouping spatially adjacent voxels into a smaller set of coherent regions, each represented by a single averaged timeseries. The choice of parcellation matters: different spatial decompositions can yield different, and sometimes inconsistent, results.

Traditional parcellations divide the brain along anatomical landmarks such as sulci and gyri. These boundaries are easy to identify and reproduce, but they do not necessarily reflect the underlying functional organisation of the brain. For a study concerned with the dynamics of neural activity, a parcellation grounded in function is preferable.

The Schaefer cortical atlas derives parcel boundaries directly from resting-state functional connectivity patterns [19]. Its gradient-weighted Markov Random Field model simultaneously groups brain locations with similar connectivity profiles and respects boundaries where those profiles change abruptly, producing spatially contiguous, functionally homogeneous parcels that align well with histological area boundaries [19].

The atlas is available at multiple resolutions. This study employs four granularities: 100, 200, 400, and 800 parcellations. Using multiple resolutions is not redundant: a finding that appears at 100 ROIs but disappears at 800 ROIs is likely an artefact of coarse spatial averaging, while one that replicates across all four scales provides much stronger evidence of a genuine effect. The seven pipeline configurations studied here (Section 4.2) are designed precisely to make this robustness testable rather than assumed.

Beyond its methodological appeal, the parcellation is consistent with the large-scale functional organisation of the cortex. When its parcels are grouped into a small number of communities, they recover the canonical functional networks that recur across resting-state studies, both at a coarse and at a finer level of description. We adopt the atlas purely as a definition of the regions of interest, but this correspondence, shown in Figure 3.1, confirms that the connectivity-derived boundaries respect the brain’s system-level architecture.

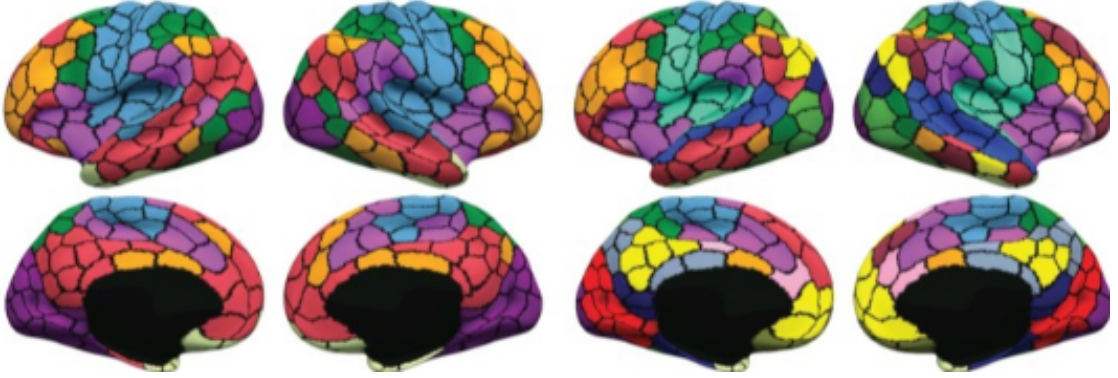


Figure 3.1: Cortical surface discretization via the multi-scale Schaefer atlas under different global constraints. The panels illustrate the geometric partition of the cerebral cortex where local, compact regions of interest (ROIs) are constrained globally by either 7 macro-scale functional networks (left) or 17 finer sub-networks (right), represented by the distinct color-coded domains. While this illustrative row displays the geometric alignment at these two reference background scales, the topological pipeline implemented in this thesis systematically evaluates higher-resolution spatial discretizations across 100, 200, 400, and 800 discrete ROIs. This multi-scale design provides a rigorous consistency check for the downstream combinatorial shape graphs generated by the Mapper algorithm. Adapted from Schaefer et al. [19].

3.3 The Emo-Film Dataset: Cohort, Stimuli, and Protocol

The Emo-FilM dataset [3] consists of fMRI recordings from 30 healthy adult participants (coded S01–S32, with S12 and S18 absent) who watched a series of short films inside the MRI scanner across four sessions of approximately two hours each. Subjects were instructed to watch the films as they would in everyday life, without any concurrent task. This naturalistic protocol is a deliberate methodological choice: films sustain continuous, ecologically valid emotional engagement over time, unlike traditional block designs in which artificial stimuli are presented in brief, isolated trials. The result is a richer and more representative sampling of the brain’s affective dynamics.

Each subject watched 14 short films and completed one 10-minute resting-state scan, during which they fixated a crosshair with eyes open. Each film run was bracketed by a 90-second washout period. The 14 films span a wide range of genres and emotional content, from animation to thriller, with durations ranging from

460 to 945 TRs (Table 3.1). This variety is intentional: comparing topological signatures across films with markedly different emotional profiles is central to the fingerprinting analysis described in Chapter 5.

Table 3.1: The 14 film stimuli in the Emo-FilM dataset including washout time [3].

Film	Genre	TRs	Duration
After the Rain	Drama	534	8:16
Between Viewings	Comedy / Drama	776	13:28
Big Buck Bunny	Animation / Comedy	528	8:10
Chatter	Thriller	464	6:45
First Bite	Romance	613	9:59
Lesson Learned	Drama	665	11:07
Payload	Drama / Sci-Fi	928	16:48
Sintel	Animation / Fantasy	710	12:02
Spaceman	Romance / Drama	744	13:25
Superhero	Drama	945	17:08
Tears of Steel	Action / Sci-Fi	607	9:48
The Secret Number	Drama	757	13:04
To Claire From Sonny	Drama	460	6:42
You Again	Romance / Drama	768	13:18

The emotional annotations for each film are stored in a serialized Python dictionary (`emotion_ts_dict.pkl`), mapping each film to a matrix of shape $T \times 50$, where T is the number of TRs and 50 is the number of annotated affective features described in Section 3.4. Once the Mapper graph is constructed, each node encodes a cluster of TR indices. Its emotional profile is obtained by averaging the corresponding rows of the annotation matrix, so that every node in the graph carries a compact affective fingerprint of the brain states it represents.

3.4 Affective Annotations: The VAD Model, the All50 Feature Set, and Temporal Alignment

Representing emotion: from discrete categories to continuous dimensions

The oldest and most intuitive way to describe emotion is through discrete categories: fear, joy, anger, sadness, and so on. This approach, associated with Ekman’s theory of basic emotions, is well suited to experimental paradigms that present isolated emotional stimuli. For naturalistic film viewing, however, it falls short: emotions

during a film evolve continuously, blend into one another, and rarely map cleanly onto a single category at any given moment.

A more flexible framework represents emotional states as points in a continuous multidimensional space. The most widely used formulation defines three orthogonal axes: Valence (pleasant to unpleasant), Arousal (calm to excited), and Dominance (submissive to dominant), collectively known as the VAD model. Any emotional state can be located within this space, and its temporal evolution becomes a continuous trajectory rather than a sequence of discrete labels.

The All50 feature set

The Emo-FilM annotation set goes beyond the VAD framework, comprising 50 continuous affective features annotated by 44 independent raters [3]. These include items from the Component Process Model of emotion, covering appraisal dimensions, motivational states, expressive and physiological components, and subjective feeling, alongside valence and arousal. The 44 raters produced continuous annotations at 1 Hz using dedicated software, and their responses were aggregated into a consensus timeseries for each feature and film.

The resulting values are raw Z -scores relative to each rater’s baseline, and they are used as such throughout this study. The signed magnitude of each feature carries meaningful information: a strongly negative valence score indicates distress, a positive one indicates joy, and zero corresponds to a neutral state. Normalising these values to a fixed $[0,1]$ window would erase this semantic baseline, making the zero point meaningless and distorting the relative intensity of affective states. All analyses therefore operate on the raw signed values, whose empirical range spans approximately -4 to $+7$.

Temporal alignment

The raw annotations are produced at 1 Hz, while the fMRI data are sampled at one volume every $TR = 1.3$ s. To align the two streams, the continuous annotation timeseries for each feature are window-averaged over each TR interval, yielding a downsampled matrix of shape $T \times 50$ that matches the fMRI timeseries sample-for-sample. This aligned matrix is stored in `emotion_ts_dict.pkl` and used in all subsequent analyses.

Chapter 4

Implementation of the Computational Pipeline

4.1 Core Architecture: From Raw BOLD Signals to Mapper Graphs

The pipeline described in Chapter 5 is implemented as three Python modules (`config.py`, `data_loader.py`, and `mapper_builder.py`) whose responsibilities map directly onto the four algorithmic steps of the Mapper framework (Section 1.5).¹ A single entry point, `build_graph()`, orchestrates the full sequence for a given subject-film-configuration triplet and returns an annotated graph together with its JSON serialisation.

Data loading and standardisation

`load_fmri()` reads the parcellated BOLD time series from disk and returns a matrix $X \in \mathbb{R}^{T \times n}$, where T is the number of TRs for the film and n is the number of Schaefer parcels at the chosen resolution. For the `.npy` format used by the 200-, 400-, and 800-parcel atlases, the first 71 TRs are discarded before the film stimulus begins to allow the haemodynamic response to stabilise. The matrix is then standardised column-wise to zero mean and unit variance, a necessary step because parcels differ substantially in baseline signal amplitude and the Manhattan distance used in the clustering step is sensitive to scale. When the target is `avg`,

¹The full implementation, including all analysis scripts and the code used to generate the figures presented in this thesis, is openly available at <https://github.com/davide-raciti/tda-brain-thesis>.

the function averages the raw matrices across all 30 valid participants before standardising, yielding the group-mean trajectory through the n -dimensional signal space.

Filter application and cover construction

`build_mapper()` applies the chosen lens function to produce the two-dimensional projection $f(X)$ used as input to the binning step (Section 1.6). The number of cover intervals is set adaptively as

$$n_{\text{intervals}} = \left\lceil \frac{T}{\rho} \right\rceil, \quad \rho = \frac{1017}{30} \approx 33.9 \text{ TR/interval}, \quad (4.1)$$

where ρ is the reference density derived from the Sintel stimulus used by Saggar et al. [1] (1017 TRs, 30 intervals), ensuring that films of different lengths produce covers of comparable topological density. The overlap fraction is fixed at 0.7 across all configurations.

Within each interval, the subset of TRs is clustered in the original n -dimensional space by `build_clusterer()`, using single-linkage agglomerative clustering with Manhattan distance. The merging threshold is not fixed but calibrated adaptively to the local geometry of X : a k -nearest-neighbour graph ($k = 3$) is fitted on X , and the 90th percentile of the resulting k -th neighbour distances is taken as the distance cutoff, following Saggar et al. [1].

Emotion annotation

Once the NetworkX graph is assembled, `annotate_nodes()` attaches an affective fingerprint to each node. The VAD annotation vectors loaded from the 3FA files assign to every TR a dominant component (Valence, Arousal, or Dominance) via a one-hot encoding of the highest-magnitude entry. For each node, the proportions of its member TRs belonging to each category are computed and stored as the node attribute `proportions`, encoding the dominant affective character of the brain state the node represents.

Output format and batch execution

Each graph is serialised to disk as a JSON file. Every node entry stores its member TR indices (`members`), cluster size, graph degree, and emotion proportions (`proportions`). Every edge entry stores the raw TR overlap count (`value`) and a normalised Jaccard weight (`strength`), defined as the overlap divided by the size of the smaller incident node.

The batch script `run_pipeline.py` iterates over all seven configurations, all 30 participants, and all 14 films, producing $30 \times 14 = 420$ subject-level graphs per configuration plus one group-average graph per film, for a total of $7 \times 434 = 3038$ JSON files.

4.2 Parameter Tuning and Multi-Scale Analysis: The Seven Configurations

A single pipeline instantiation is defined by two independent choices: the spatial resolution of the brain parcellation and the dimensionality reduction technique used as the Mapper lens. Rather than committing to a single parameter set, this study systematically explores a grid of configurations designed to test whether the observed topological findings are genuine neurobiological properties of the data or artefacts of a particular algorithmic choice.

The configuration grid

Four parcellation resolutions from the Schaefer atlas are considered: 100, 200, 400, and 800 ROIs. Two lens functions are evaluated: Reciprocal Isomap (**riso**) and t-SNE (**tsne**). The full factorial combination yields eight pairings, of which seven are retained: the `800_tsne` configuration is excluded on computational grounds, as t-SNE scales poorly to the embedding of 800-dimensional vectors over long film runs. The seven configurations are listed in Table 4.1.

Table 4.1: The seven pipeline configurations explored in this study. The primary configuration used for all main analyses is `100_riso`.

Label	Resolution (ROIs)	Lens
<code>100_riso</code>	100	Reciprocal Isomap
<code>100_tsne</code>	100	t-SNE
<code>200_riso</code>	200	Reciprocal Isomap
<code>200_tsne</code>	200	t-SNE
<code>400_riso</code>	400	Reciprocal Isomap
<code>400_tsne</code>	400	t-SNE
<code>800_riso</code>	800	Reciprocal Isomap

The spatial resolution axis

Varying the parcellation resolution tests whether the topological findings replicate across spatial scales. A pattern that emerges at 100 ROIs but dissolves at 800 ROIs is likely driven by coarse spatial averaging that collapses distinct functional territories into a single parcel signal, rather than by genuine structure in the underlying neural dynamics. Conversely, a finding that is stable across all four resolutions provides strong evidence of a scale-invariant neurobiological effect. The four resolutions span nearly an order of magnitude in spatial granularity, from the coarsest functional parcellation commonly used in the literature to a fine-grained decomposition that approaches the scale of individual cortical areas.

The lens axis

The mathematical properties of the two lens functions are described in detail in Section 1.6. In the context of the configuration grid, what matters is the distinct geometric commitment each lens imposes on the Mapper output.

Reciprocal Isomap organises the temporal points according to their geodesic distances along the data manifold, preserving global continuity. t-SNE, by contrast, optimises for local neighbourhood separation at the expense of global distances, accentuating cluster boundaries and suppressing the continuous flow between brain states. The two lenses therefore probe complementary aspects of the fMRI trajectory: a finding that survives both projections is unlikely to be an artefact of the embedding geometry.

The primary configuration

All main results reported in this thesis are computed on the `100_riso` configuration; the remaining six serve as robustness checks. The choice of Reciprocal Isomap as the primary lens follows directly from the systematic lens comparison reported in Geniesse et al. [20], where Reciprocal Isomap was found to best capture both local and global data structure in BOLD time series, outperforming linear methods (PCA), standard Isomap, UMAP, and t-SNE on a naturalistic fMRI dataset closely analogous to the one analysed here.

The choice of 100 ROIs as the primary resolution reflects a different consideration. Several downstream analyses, in particular community detection via the Louvain algorithm (Section 5.3) and graph portrait divergence (Section 5.5), operate directly on the Mapper graph structure rather than on the brain parcellation. At coarser resolutions the Mapper graphs are smaller and more tractable, making the 100-ROI configuration the natural reference for computationally intensive cross-subject comparisons. Robustness across the higher-resolution configurations is verified separately for each analysis.

Figure 4.1 shows one such 100_riso graph, with nodes coloured by mean acquisition time, illustrating how the temporal trajectory of the scan folds into the topology of the shape graph.

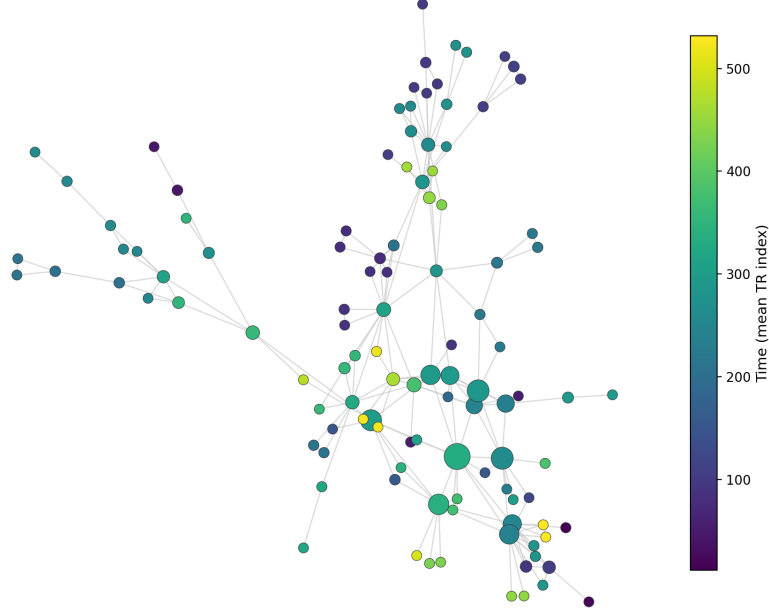


Figure 4.1: Qualitative topology and temporal progression of a Mapper shape graph. The visualization presents the largest connected component of a single empirical graph (film *Sintel*, subject S13, configuration 100_riso). Each node is color-coded according to the mean fMRI timeframe (TR) of the scan volumes it encompasses, and scaled proportionally to the total number of frames contained within the corresponding cluster. The temporal trajectory of the neuroimaging session systematically folds into the geometric shape of the graph: elongated peripheral arms encapsulate consecutive, linear timepoints, whereas the highly dense central core aggregates brain states that recur at distinct moments throughout the scan, yielding the highly branched topological structure that the Mapper algorithm is designed to expose.

4.3 Topological Node Classification

Each node of a shape graph carries a topological role, determined by how it sits within the network. Crossing the coreness and normalised betweenness centrality

introduced in Section 1.2 assigns each node to one of four categories, used throughout the meso-scale and micro-scale analyses.

Each node i is classified as:

- **Peripheral:** $C_i^B = 0$ and $c_i \leq 1$. Lies on no geodesic and belongs at most to the 1-core; a weakly integrated state.
- **Bridge:** $C_i^B > 0$ and $c_i \leq 1$. Mediates paths between other nodes but is not part of a dense core; a local connector.
- **Connector Hub:** $C_i^B > 0$ and $c_i \geq 2$. Both a routing bridge and a member of the dense core, combining global reach with local cohesion.
- **Provincial Hub:** $C_i^B = 0$ and $c_i \geq 2$. A core member that lies on no shortest path, locally central but with no bridging role (for example a node inside a clique or triangle).

This classification lets the micro-scale analysis of Section 5.4 test whether a node's topological role is linked to the emotional activation and complexity of the corresponding brain state. Figure 4.2 maps the four categories onto an example shape graph for the film *Sintel*.

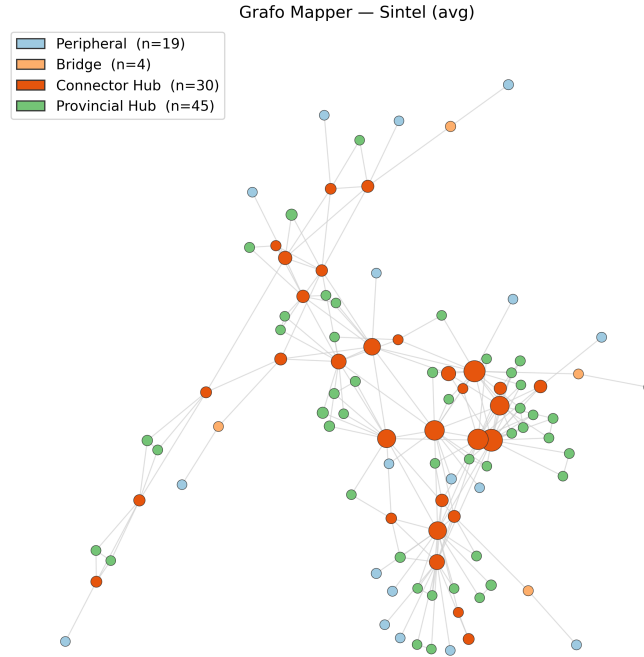


Figure 4.2: Topological node classification mapped onto the empirical shape graph of the film *Sintel* (average configuration). Nodes are color-coded according to the four functional categories derived from the meso-scale metrics: Peripheral ($n = 19$, light blue), Bridge ($n = 4$, light orange), Connector Hub ($n = 30$, dark orange), and Provincial Hub ($n = 45$, green). The spatial arrangement highlights the concentration of connector hubs within the integrative backbone of the network, contrasting with the local nesting of provincial hubs and the expansion of peripheral configurations towards the boundaries of the topological landscape.

4.4 Feature Propagation: Mapping Raw Affective Annotations onto Graph Nodes

The VAD annotation described in Section 4.1 assigns each node a dominant emotional category for visualisation purposes. The analysis developed in Chapter ?? requires a richer, quantitative description of each node’s affective content. This section describes how the full 50-dimensional CPM annotation matrix introduced in Section 3.4 is projected onto the graph structure, and how two node-level metrics are derived from it.

Node-level affective vectors

For each node in a Mapper graph, the set of member TR indices is extracted from the JSON output. The rows of the within-film All50 matrix corresponding to those indices are retrieved from `emotion_ts_dict.pkl` and averaged across members, yielding a single vector $\mathbf{e} \in \mathbb{R}^{50}$ that represents the mean affective profile of the brain state encoded by that node. All subsequent metrics are computed from $\mathbf{e}$.

Emotional Activation

Emotional Activation quantifies the fraction of CPM features that are significantly engaged in a given node. The All50 matrix is first Z -scored within each film independently:

$$z_{t,j} = \frac{a_{t,j} - \mu_j}{\sigma_j}, \quad (4.2)$$

where μ_j and σ_j are the mean and standard deviation of feature j across all T TRs of that film. Within-film standardisation is adopted because it captures emotional peaks relative to the specific affective baseline of each stimulus, rather than against a global norm that would conflate films with inherently different affective ranges. The node-level Z -scored profile $\bar{\mathbf{z}} \in \mathbb{R}^{50}$ is the mean of the Z -scored rows of the member TRs. Activation is then defined as:

$$A = \frac{1}{50} \sum_{j=1}^{50} \mathbf{1} [|\bar{z}_j| > 1.5]. \quad (4.3)$$

A node with high Activation concentrates TRs during which a large proportion of the emotional response components deviate substantially from the film's own baseline.

Emotional Complexity

Emotional Complexity measures the distributional diversity of the affective profile $\mathbf{e}$. The raw mean profile is shifted to be non-negative and normalised to a probability simplex:

$$p_j = \frac{e_j - \min_k e_k}{\sum_k (e_k - \min_k e_k)}, \quad (4.4)$$

and Shannon entropy is computed in bits:

$$H = - \sum_{j=1}^{50} p_j \log_2 p_j. \quad (4.5)$$

Raw entropy values are retained without further normalisation. A normalised entropy would express diversity relative to the theoretical maximum for 50 features, collapsing the absolute difference between a node with two dominant features and one with twenty, which carries genuine neurobiological meaning. High Complexity therefore indicates that a node's member TRs span a broad and evenly distributed range of affective components, whereas low Complexity indicates concentration around a small number of features.

Chapter 5

Topological and Geometric Analysis: Results and Discussion

5.1 Emotional Characterization of the Film Stimuli

Before turning to the brain networks, we characterise the affective content of the fourteen film stimuli directly from the All50 annotation matrices (Section 3.4). This establishes the emotional ground truth against which the topological findings of the following sections are interpreted. All quantities in this section are produced by `scripts/5_1_emotional_profile.py`.

Emotional variability ranking

For each film we measure how much its emotional content varies over time. For each of the seven discrete emotions we build a ten-bin histogram of its values across the film's TRs and take the Shannon entropy of that histogram; the film's score is the mean entropy over the seven emotions. A film whose emotions sweep through many intensity levels over time scores high, while one that stays in a narrow emotional band scores low. This temporal-variability measure is distinct from the node-level Emotional Complexity of Section 4.4, which is the entropy of a single averaged profile; the two answer different questions. The ranking runs from *Payload* at the top (≈ 2.87 bits) to *To Claire From Sonny* at the bottom (≈ 2.17 bits), confirming that the stimuli cover a graded spectrum of emotional dynamism rather than a single affective register.

Affective signatures

While the entropy ranking summarises emotional variability in a single scalar, it deliberately discards the identity of the emotions involved: two films with the same entropy can be driven by entirely different feelings. To recover this qualitative dimension we plot, for each film, a radar chart over the seven discrete emotion axes of the All50 set (Figure 5.1), where every axis reports the mean response across the timepoints of the film on a shared radial scale.

Read this way, each film traces a distinct affective signature, and the three examples in Figure 5.1 span the range of the entropy ranking. *Sintel*, near the top of that ranking, spreads almost evenly over the axes with no single emotion dominating. *Chatter* is pulled toward surprise and, to a lesser extent, fear. *ToClaireFromSonny*, the lowest-entropy film of the set, traces the most lopsided profile, suppressed on anger, guilt and sadness and weighted toward disgust and fear. That three short films already occupy such different regions of the emotion space confirms that the stimuli are emotionally differentiated rather than interchangeable, a precondition for any analysis that seeks to relate brain topology to affective content.

Intra-film emotional coherence

Finally, `emotion_coherence.csv` reports, for each film, the mean pairwise Pearson correlation among the seven discrete emotion timeseries. This measures how tightly the affective dimensions co-vary over the course of the film: a high value denotes a stimulus whose emotions rise and fall in concert, a low value a more multidimensional emotional trajectory.

It is worth stressing a structural property of the dataset here: the All50 annotations are film-level, obtained as a consensus over 44 raters and shared across all subjects. Consequently, no genuine *inter-subject* emotional alignment can be measured from these annotations, since the emotional timecourse is identical for every viewer of a given film by construction. The claim that naturalistic cinematic stimuli act as a collective driver, synchronising responses across the cohort, is therefore tested not on the annotations but on the brain data, where it emerges from the geometric fingerprinting analysis of Section 5.5.

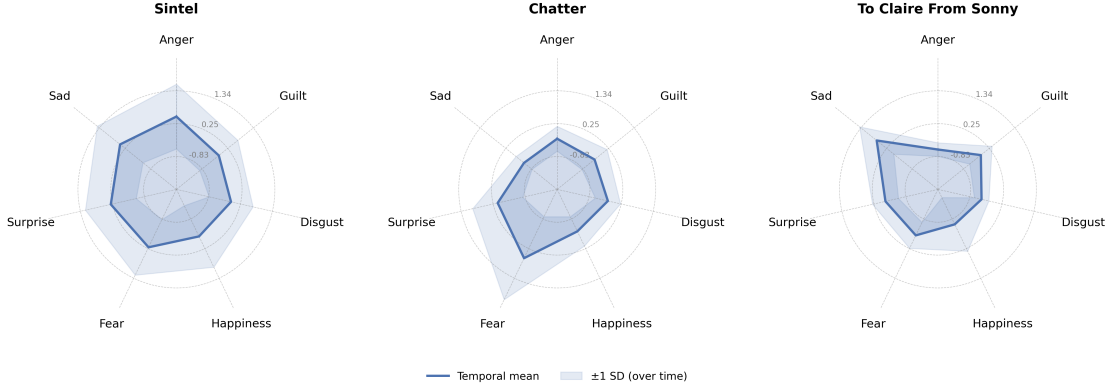


Figure 5.1: Affective signatures of three example films. Radar charts over the seven discrete emotion axes of the All50 set; the bold line represents the mean response across the timepoints of the film and the shaded band indicates its ± 1 standard deviation over time, on a radial scale shared across panels. *Left: Sintel* spreads almost evenly over the axes with no single emotion dominating; *Center: Chatter* is visibly pulled toward fear, the only axis clearly above baseline; *Right: To Claire From Sonny*, the lowest-entropy film of the dataset, exhibits a highly lopsided profile weighted heavily toward sadness.

5.2 Macro-Scale Analysis: Global Topology, Rest Baseline, and Null Model Validation

We begin at the coarsest scale, asking whether the global organisation of the Mapper networks is reshaped by narrative stimulation and whether it departs from random connectivity. The metrics in this section are computed over the 420 film networks of the primary configuration (100_riso) by `scripts/5_2a_global_topology.py` and `scripts/5_2b_power_law.py`.

Global topography

Four normalised descriptors summarise each network: density, largest connected component (LCC) ratio, diameter ratio, and maximum degree. Their distributions across the 420 graphs are stable and tightly concentrated, providing a reference baseline for the comparisons that follow. The most salient feature is the LCC ratio, whose mean of ≈ 0.53 indicates that on average only about half of the nodes belong to the giant component. This structural fragmentation is addressed at the meso scale (Section 5.3).

Rest versus Film

To test whether emotional, narrative stimulation alters the global network boundaries, we compare each subject’s mean film network against their resting-state network with a two-sided paired Wilcoxon signed-rank test. Across all all four macro-metrics the tests are non-significant ($p > 0.05$). The macro-scale shape of the brain network is therefore statistically invariant between Rest and Film: continuous emotional engagement does not redraw the global topological envelope of brain dynamics. This null result is not a weakness but a baseline, since it localises any stimulus-driven reorganisation to finer scales, which the following sections probe directly.

Degree distribution and power-law modelling

We next examine the pooled degree sequence of all film networks under the Clauset–Shalizi–Newman framework (Section 2.2.3). Maximum-likelihood fitting yields a power-law exponent $\alpha = 2.412 \pm 0.009$, comfortably within the range $2 < \alpha < 3$ characteristic of biological and physical networks. A bootstrap goodness-of-fit test nonetheless rejects the pure power-law hypothesis ($p = 0.000$), motivating the model-selection comparison. Taking the pure power law as reference in the Vuong likelihood ratio test, the exponential model is decisively rejected in its favour ($\mathcal{R} = +21.49$), while the truncated power law in turn significantly outperforms the pure power law ($\mathcal{R} = -22.87$). The degree distribution is thus best described by a truncated power law: heavy-tailed, but with an exponential cutoff at high degree.

This cutoff is structurally meaningful. A pure scale-free tail would permit arbitrarily large hubs, whereas the empirical right-tail truncation is the signature of a physical network embedded in space and constrained by finite metabolic resources. The cost of sustaining very high-degree integrator states bounds the maximum degree, exactly as observed in other energy-limited biological networks. Figure 5.2 shows the pooled degree distribution as a complementary cumulative distribution function, with the maximum-likelihood power-law and truncated power-law fits.

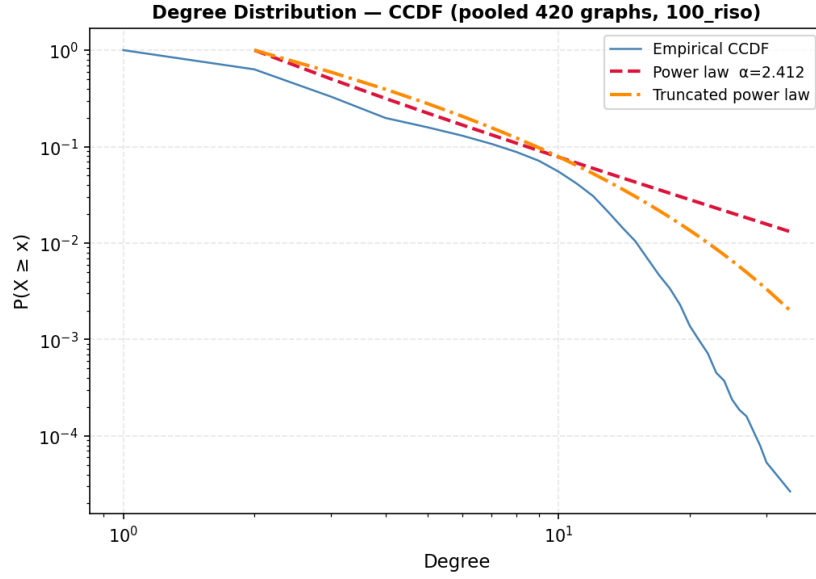


Figure 5.2: Degree distribution. Complementary cumulative distribution function (CCDF) of the pooled node degrees ($n = 37,505$, 420 graphs, 100_riso) on log–log axes, with the maximum-likelihood power-law (dashed) and truncated power-law (dash-dotted) fits. The tail is heavy and well described by an exponent $\alpha = 2.41$; a likelihood-ratio test favours the truncated power law over the pure power law ($p < 0.001$), pointing to an exponential cutoff at high degree.

Null-model validation

Finally, we situate this structure against standard random-graph baselines. For each network we generate Erdős–Rényi (ER) and Barabási–Albert (BA) graphs matched in size and edge density, and compare the empirical LCC ratio, diameter ratio, and maximum degree against the nulls via paired Wilcoxon tests.

The empirical networks differ significantly from both baselines on all three descriptors (all $p < 0.001$). They are markedly more fragmented than either null, and their hubs are intermediate: larger than the homogeneous ER degree but well below the runaway hubs of BA, consistent with the bounded, truncated heavy tail reported above. Neither homogeneous (ER) nor scale-free (BA) random connectivity reproduces this combination of fragmentation, heavy tails, and bounded hubs. Figure 5.3 compares the three global descriptors against the substrate-matched ER and BA null models.

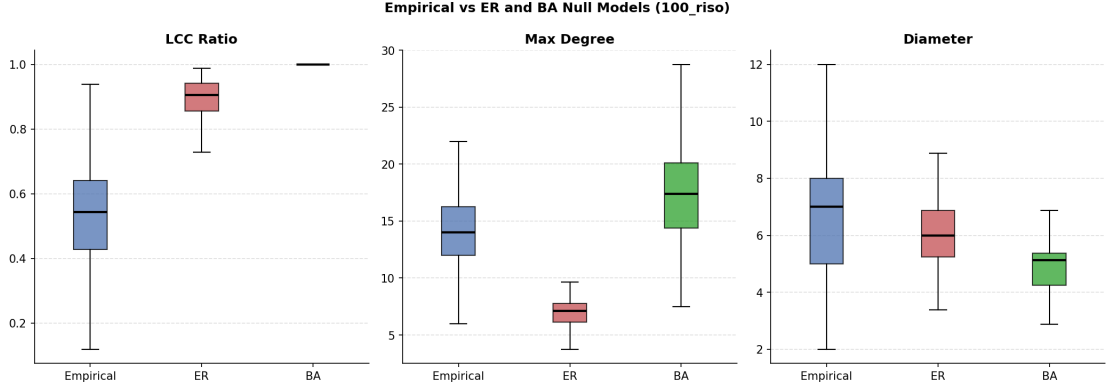


Figure 5.3: Comparison with null models. Three global metrics (LCC ratio, maximum degree, diameter) for the empirical film graphs against substrate-matched Erdős-Rényi (ER) and Barabási-Albert (BA) null models, computed over the 420 graphs (100_riso).

5.3 Meso-Scale Analysis: K-core, Betweenness, Community Structure, and Cross-Resolution Consistency

The macro scale revealed an invariant global envelope. We now descend to the meso scale, where the internal organisation of the networks proves far more responsive to narrative stimulation. The metrics applied here, k-core decomposition, betweenness centrality, and Louvain modularity, are defined in Sections 1.2 and 1.3; this section reports their empirical behaviour. All results are produced by `scripts/5_3a_kcore_betweenness.py` and `scripts/5_3b_community.py` on the 100_riso configuration, with the resting-state networks as baseline.

K-core decomposition

Across the empirical Mapper graphs the node coreness spans a narrow interval from 0 to 4, with a mean maximum coreness of 3.25, consistent with the shallow hierarchical depth expected from the sparse, locally clustered output of the Mapper algorithm. The decomposition is nonetheless sensitive to condition. While the mean core number is statistically unchanged between Rest and Film, the depth of the innermost core (`max_core`) is significantly higher during Film ($p = 0.020$, two-sided paired Wilcoxon). Narrative processing therefore drives the emergence of a deeper, more tightly integrated central core, a routing backbone that the resting brain does not sustain.

Betweenness centrality

Betweenness centrality is heavily zero-inflated in these graphs: only 22.8% of nodes lie on at least one geodesic and thus carry non-zero betweenness, reflecting the locally clustered topology in which most nodes sit inside dense neighbourhoods or on peripheral leaves.

The maximum betweenness does not separate the two conditions, but the distribution does shift: the mean betweenness decreases slightly under the Film condition ($p = 0.035$), while the Gini coefficient of the betweenness distribution rises ($p = 0.014$).

Together, these results point to a redistribution rather than an amplification of the bridging role: under narrative stimulation, the geodesic traffic concentrates on a smaller, more unequal set of bridge nodes, even as the average bridging load itself edges down.

Community structure on the giant component

Community detection is run on the largest connected component, which holds about half of the nodes (Section 5.2), since modularity is not well defined across the many small peripheral components of a Mapper graph: those components would be counted as separate communities, inflating the partition with what are really isolated fragments rather than genuine modules. We compare the empirical partition against Erdős–Rényi and Barabási–Albert nulls matched to the same component. The empirical core is significantly more modular than either null: its modularity $Q \approx 0.508$ exceeds the ER value ≈ 0.419 and the BA value ≈ 0.402 (both $p < 0.001$). At the same time it splits into fewer communities, around 5.34 on average, against 5.90 for ER and 6.07 for BA ($p < 0.001$). The connected core of the brain network therefore carries genuine modular structure that neither random wiring nor scale-free growth reproduces, and it is organised into a small number of large, cohesive modules rather than many smaller ones. Figure 5.4 renders all three mesoscale descriptors on a single example shape graph.

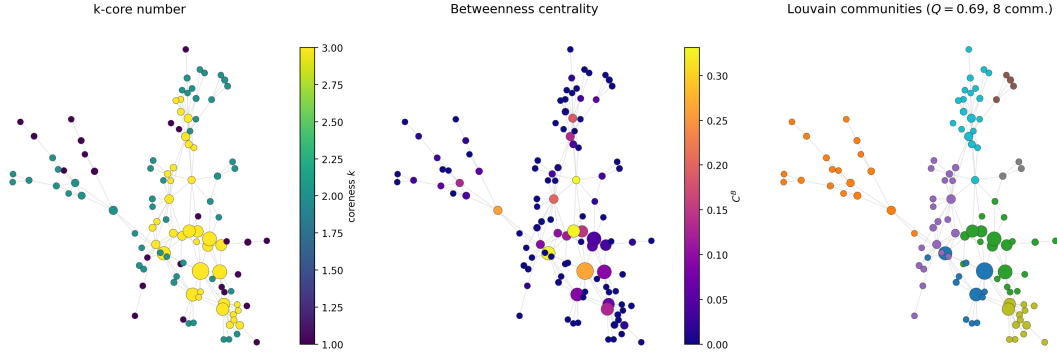


Figure 5.4: Mesoscale structure on an example shape graph. The same empirical Mapper graph (*Sintel*, subject S13, 100_riso; largest connected component shown) is presented under a shared spatial layout, color-coded by three complementary mesoscale descriptors. *Left:* k -core decomposition, with the innermost core ($k = 3$) concentrated within the dense central backbone, while the outer branches form the 1-shell periphery. *Middle:* betweenness centrality, which selectively isolates the strategic bridge nodes joining the peripheral arms to the central core. *Right:* Louvain community detection ($Q = 0.69$, 8 modules), which clusters the topological arms into distinct, well-defined functional modules. Across all panels, node size is proportional to the number of fMRI timepoints encompassed within each cluster.

Rest versus Film and per-pair reconfiguration

Comparing community structure across conditions, the LCC modularity Q shows no significant Rest–Film difference ($p = 0.715$), whereas the number of communities shifts significantly ($p = 0.015$). The modular partition is thus reorganised in granularity without changing its overall quality. A finer per-pair analysis over all 91 film pairs, with Benjamini–Hochberg and Bonferroni corrections for the many simultaneous tests, confirms that this reconfiguration is content-dependent: 66 of the 91 pairs remain significant after FDR correction, and 38 under the stricter Bonferroni control, with heterogeneous directions that reflect the distinct narrative structure of each stimulus rather than a single global trend. Figure 5.5 shows the paired Rest–Film comparison of the three mesoscale descriptors across the 30 subjects.

Cross-resolution consistency

Finally, we test whether these descriptors are robust to the choice of parcellation scale by correlating the metrics computed at 100 ROIs against the higher-resolution riso configurations, up to 800 ROIs. The two families behave very differently.

Betweenness centrality is scale-invariant, with a strong rank correlation across resolutions ($r \approx 0.71$), and therefore constitutes a reliable topological marker. The k-core decomposition, by contrast, is highly scale-dependent, its rank correlation across resolutions remaining weak, so coreness-based claims hold only at the resolution at which they are measured. This dissociation is itself a methodological finding, separating the robust descriptors of the pipeline from the fragile ones.

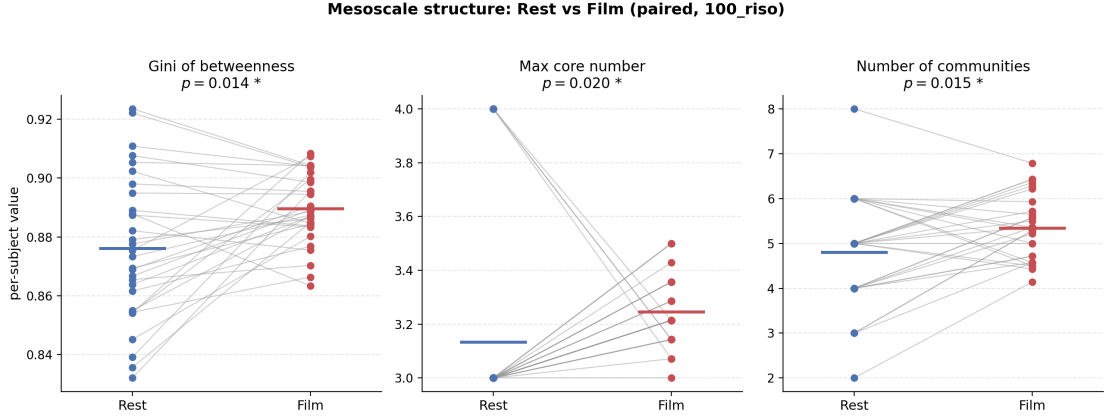


Figure 5.5: Mesoscale structure: Rest vs. Film. Paired comparison of three mesoscale descriptors across the 30 subjects (100_riso). For each subject, the film value (red) represents the mean over that subject’s cinematic scans, paired with the same subject’s resting-state baseline value (blue); thin lines join the paired observations, while thick horizontal bars mark the respective group means. All three descriptors exhibit a significant increase from rest to film viewing (paired Wilcoxon signed-rank test): the Gini coefficient of betweenness centrality ($p = 0.014$), the maximum core number ($p = 0.020$), and the absolute number of Louvain communities ($p = 0.015$). Cinematic stimulation thus sharpens the mesoscale organisation, yielding deeper hierarchical cores, a more concentrated bridging infrastructure, and a finer division into a larger number of communities (at unchanged overall modularity Q) relative to resting state.

5.4 Micro-Scale Analysis: Nodal Centrality and Emotional Profile

Having established how the networks reorganise at the meso scale, we now ask the question at the heart of this thesis: is the topological role of a single node related to the emotional content of the brain state it represents? The analysis is conducted at the level of individual nodes, pooling all graphs of the 100_riso configuration, and is produced by `scripts/5_4_nodal_emotion.py`.

Setup

Pooling the nodes of the 420 empirical networks yields a dataset of roughly 44,000 independent nodes. Each node carries its three topological descriptors (core number, degree, betweenness centrality), its functional category as defined in Section 4.3, and its two emotional metrics, Emotional Activation and Emotional Complexity, computed from the All50 annotations as described in Section 4.4.

Topological centrality versus emotional activation

The Spearman rank correlations between the three topological predictors and Emotional Activation are all negative and highly significant: core number $r = -0.324$, degree $r = -0.364$, and betweenness centrality $r = -0.455$, the strongest of the three (all $p < 0.001$). The more central a node is in the network topology, the lower the emotional activation of the brain state it encodes. The correlations with Emotional Complexity, although also significant at this sample size, are an order of magnitude smaller and negligible in practice ($|r| < 0.03$): topological centrality tracks the *intensity* of a node’s affective state, not the *diversity* of its profile.

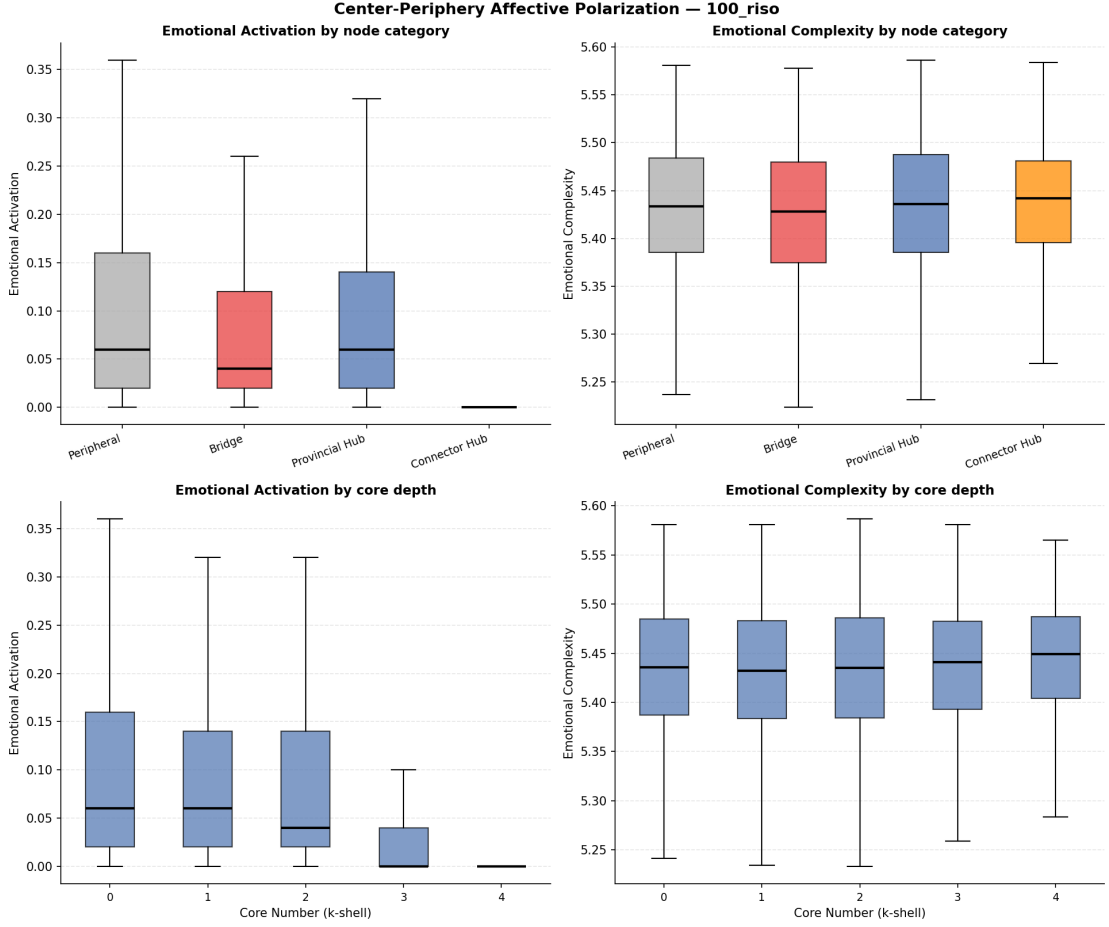


Figure 5.6: Nodal topology versus emotional profile. Emotional Activation (left column) and Emotional Complexity (right column) of the $\approx 44,000$ pooled nodes (100_riso), grouped by node category (top row) and by core depth (k -shell, bottom row); boxes show the median and interquartile range. Activation falls sharply for the most integrated nodes: it is lowest in the Connector Hubs (mean 0.025 against ≈ 0.11 for the other categories) and decreases monotonically with core depth, nearly vanishing in the innermost shell. Emotional Complexity, by contrast, is essentially flat across every category and every core level (means within 0.01 of one another), indicating that the topological organisation tracks the intensity of the affective state but not the diversity of its profile.

Emotional profile by node category

The four-way classification refines this picture and locates the effect precisely. Mean Emotional Activation is by far the lowest in Connector Hubs (0.025), the nodes that are at once deep in the core and act as bridges. It is intermediate in Bridges

(0.098) and high in Provincial Hubs (0.113), which are core members but lie on no shortest path, almost matching the peripheral nodes (0.116). Core membership alone therefore does not bring low activation: Provincial Hubs sit in the dense core yet pattern with the periphery. What marks the emotionally neutral states is the connector-hub role specifically, the combination of deep integration and a bridging position, consistent with betweenness being the strongest of the three predictors. A Kruskal-Wallis test confirms that Activation differs across the four categories ($p < 0.001$); the differences in Complexity, though statistically detectable given the large sample, are negligible in magnitude, with category means all within 5.42 to 5.43.

Interpretation

Taken together, these results describe a clear dissociation between topology and emotion, visualised in the centerpiece figure (Figure 5.6), which crosses both emotional metrics against node category and core depth. The emotionally neutral, baseline brain configurations are concentrated in the connector hubs, the nodes that are both deeply integrated and serve as bridges between regions of the graph. These are the frequently visited transit states through which the dynamics routinely pass. Moments of strong emotional activation, where many affective features deviate sharply from the film’s baseline, are pushed away from this bridging core toward the low-degree periphery, and also toward the provincial hubs, which sit in the core but carry no bridging role. The picture is therefore not a simple centre-versus-periphery split but a sharper one: emotion is suppressed precisely where integration and bridging coincide, and present everywhere else. This is consistent with betweenness, the measure of bridging, being the strongest topological correlate of emotional activation.

5.5 Geometric Fingerprinting: Film Entrainment vs. Individual Signatures via GPD

The preceding sections compared scalar metrics. We close the results with a holistic comparison of network shape itself, asking whether the geometry of a subject’s Mapper graph is governed more by the film they are watching or by their individual identity. The comparison is performed with the Network Portrait Divergence (Section 1.4), the scale-free, information-theoretic distance $D_{JS} \in [0, 1]$ between two graphs. All results are produced by `scripts/5_5_gpd_fingerprint.py`.

Film entrainment at the global scale

We contrast two families of distances. The within-subject distance compares the same subject across different films, measuring how much one brain’s topology changes with the stimulus. The within-film distance compares different subjects watching the same film, measuring how much brains differ from one another under a shared stimulus. A Mann-Whitney U test shows that within-subject distances ($D_{JS} \approx 0.355$) are significantly larger than within-film distances ($D_{JS} \approx 0.331$), giving an entrainment ratio of ≈ 1.073 ($p < 10^{-32}$). The interpretation is direct: the same brain changes its geometric shape more when the film changes than two different brains do while watching the same film. The external cinematic stimulus therefore acts as a powerful topological driver, overriding individual signatures and entraining different brains toward a common network geometry during the same movie. This is the topological counterpart of the collective emotional alignment that could not be measured on the film-level annotations of Section 5.1.

Individual fingerprinting at the fine scale

Entrainment does not preclude a residual individual signature. To detect it we build, for each of the 91 film pairs, a 30×30 subject-by-subject distance matrix whose entry (i, j) is the divergence between subject i in the first film and subject j in the second. A persistent fingerprint would make the diagonal, the same subject across the two films, systematically smaller than the off-diagonal. In 83 of the 91 film pairs (91%) the diagonal is on average smaller than the off-diagonal, a consistent direction that the per-pair Wilcoxon test confirms as significant in only 20 of them, even though the pooled Mann–Whitney test over all matrices is highly significant. Quantified as an identification rate, the fraction of subjects whose nearest neighbour in the second film is themselves, the mean rate of ≈ 0.049 exceeds the random chance baseline of $1/30 \approx 0.033$. Figure 5.7 presents the aggregated subject-by-subject and film-by-film divergence matrices.

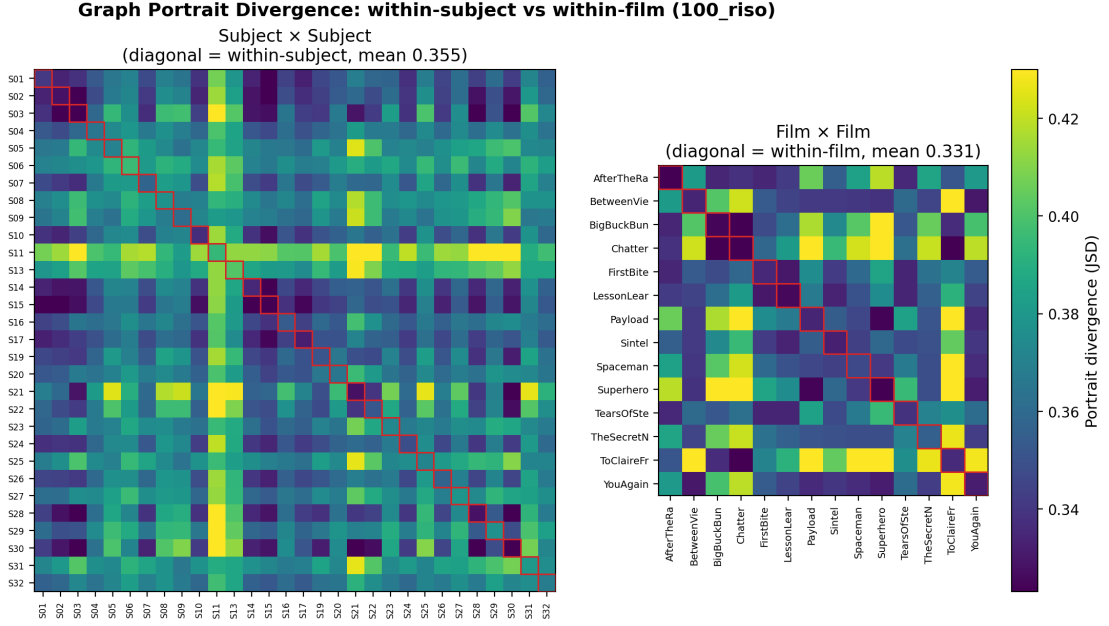


Figure 5.7: Within-subject versus within-film fingerprint. Aggregated Graph Portrait Divergence (based on Jensen-Shannon Divergence, JSD) between the 420 Mapper graphs (`100_riso`), presented as a subject $\times$ subject matrix (left, 30 subjects) and a film $\times$ film matrix (right, 14 films) on a shared colour scale. In the subject matrix, each cell averages the divergence over all pairs of distinct films, meaning its diagonal captures the within-subject divergence (same subject across different films, mean 0.355). In the film matrix, each cell averages over all pairs of distinct subjects, meaning its diagonal captures the within-film divergence (same film across different subjects, mean 0.331). The film diagonal is visibly darker than its corresponding off-diagonal elements, whereas the subject diagonal barely stands out. This asymmetry (within-film $<$ within-subject, ratio 1.073, Mann–Whitney $p < 0.001$) indicates that the external stimulus, rather than individual identity, is the dominant driver of Mapper-graph topology, leaving only a weak residual individual fingerprint (identification rate 0.049 versus 0.033 chance baseline).

Synthesis

The two analyses are complementary rather than contradictory. At the global scale the film stimulus dominates, forcing different brains into a shared geometric shape. Beneath that shared shape, a subtle but systematic individual fingerprint persists, weak in absolute terms yet significantly above chance. The topological geometry of brain dynamics is thus primarily stimulus-driven, while still retaining a faint trace of who is watching.

Chapter 6

Conclusions

6.1 Summary of Findings

This thesis asked a single question: do the Mapper graphs built from whole-brain fMRI trajectories carry information about the emotional content of a film and about the identity of the person watching it? The different analyses presented in Chapter 5 investigate this question across complementary scales and analytical lenses, revealing that the answer depends on the level at which the system is examined. Emotion and identity are not uniformly represented across the network. Rather, they appear at specific structural scales and in specific parts of the graph.

At the **macro scale** (Section 5.2), the global structure remains the same. None of the four global metrics separated Film from Rest, suggesting that the large-scale shape of the brain mapper graph does not change when the subject watches a film. This does not mean the networks lack structural complexity. Their degree distribution are consistent with a truncated power law, with an exponential cutoff that may reflect constraints on the maintenance of large hubs. The graph structure also differ from Erdős–Rényi random graphs, matched for size and density. Thus, the mapper graphs exhibit non-random organization, while their overall global shape remains relatively stable.

Under this stable global organization, structural changes emerge at the **meso-scale** level (Section 5.3). During film viewing, the innermost k -core becomes deeper, the distribution of betweenness centrality changes, and the number of communities within the giant component varies. In contrast, the global modularity remains largely unchanged, likely because the giant component comprises only a small fraction of the full graph. The cross-resolution analysis also showed which metrics are consistent across parcellation scales: betweenness centrality is relatively stable across parcellation scales, whereas k -core showed substantial scale dependence.

At the **micro scale** (Section 5.4), the link between topology and emotion

becomes more visible. Over about 44,000 nodes, every measure of topological centrality was negatively correlated with Emotional Activation and showed no correlation with Emotional Complexity. That is, the central core of the Mapper graph links frequently visited states that are emotionally heterogeneous, but predominantly neutral on average. In contrast, states associated with high emotional activation were more likely to occur in peripheral regions of the graph, characterized by low degree and low coreness. Overall, these results suggest that emotional activation is preferentially associated with topologically peripheral rather than central brain states. This is consistent with existing work on resting-state fMRI analyses [8].

At the level of **whole-network comparison** (Section 5.5), the Network Portrait Divergence, a network similarity measure, showed that the film drives a shared network shape across subjects. Distances between different subjects watching the same film are smaller than distances of the same subject across different films, so the stimulus pulls different brains toward a common network shape. Even so, a weak but consistent individual signature remains: each subject can be matched to themselves across films at a rate above chance.

The main result of this work can be stated simply. The geometry of brain dynamics encodes emotion not in its global structure but at the meso and micro scales, and most clearly in its peripheral, short-lived states. At the same time, the film acts as the main driver of network shape across subjects, without fully removing the individual signature of the viewer.

6.2 Methodological Limitations

The findings above should be read together with a number of limitations.

Network fragmentation. On average only about half of the nodes lie in the giant component (mean LCC ratio ≈ 0.53 , Section 5.2). This means that nearly half of the nodes sit in small, disconnected pieces of the graph. Many standard network metrics assume a connected graph, so they had to be restricted to the giant component, as was done for the community analysis. This limits how directly some global measures can be applied.

Sample size. The dataset contains 30 subjects. This is enough to find group-level effects, but it is small for the biometric part of the work. The individual fingerprint is significant, yet weak in absolute terms, and a larger cohort would be needed to identify single subjects with high precision.

Sensitivity to parcellation scale. The cross-resolution analysis (Section 5.3) showed that not all metrics are stable across spatial scales. Betweenness centrality is robust, but k -core decomposition changes substantially between 100 and 800 ROIs. Any result based on k -core therefore holds only at the resolution where it

was measured, and should not be read as scale-independent.

Granularity of the emotional annotations. The All50 annotations are defined at the film level. They are a single consensus over 44 raters and are the same for every subject who watches a given film. Because of this, the emotional analyses cannot separate the personal emotional experience of one subject from the shared, stimulus-driven signal. Emotion can be linked to topology only through the common emotional content of the film, not through individual affective responses.

6.3 Future Perspectives

Several directions follow naturally from this work, and some of them address the limitations listed above.

From hand-crafted metrics to learned representations. This thesis described the Mapper graphs through a fixed set of topological metrics chosen in advance. A natural next step is to let the representation be learned from the data instead. Graph Neural Networks can be trained directly on the Mapper graphs, using the multi-resolution exports already produced by the pipeline, to learn compact embeddings of network shape. Such embeddings could then be used for emotional decoding or for subject identification, and might capture structure that fixed metrics miss. A self-supervised setup would be a good fit, since it does not need large amounts of labelled data.

Larger datasets. The main limit on the biometric results is the size of the cohort. Applying the same pipeline to large, open neuroimaging datasets would give more statistical power and would test whether the present findings hold beyond this specific group of 30 subjects. The pipeline is already automated and configuration-driven, so scaling it to more subjects and films is mostly a matter of compute.

Subject-level emotional data. The emotional analyses were limited by the film-level annotations. Pairing the topological pipeline with emotional reports collected from each subject, where such data are available, would make it possible to separate the shared stimulus-driven signal from the individual emotional experience. This would allow a stronger test of whether brain topology tracks how a specific person feels, and not only the emotional content of the film.

Code Availability

The complete source code supporting this thesis, comprising the computational pipeline, all analysis scripts, and the code used to generate the figures, is openly available at <https://github.com/davide-raciti/tda-brain-thesis>.

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